

Technical note: searching the ASTRA quantile data vectors for the largest Fisher gain over the full-sample 2PCF

ASTRA clustering project

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Abstract

Using the complete grid of full-box runs on disk (nine AbacusSummit cosmologies $\times$ 50 HOD draws), we ask which ASTRA environment-split statistic, or which pair of statistics, extracts the most cosmological Fisher information beyond the plain full-sample auto-correlation. We sweep all 16 single quantile vectors and all 120 pairs, in two framings (standalone and added to the full auto), scoring each by the HOD-marginalised Fisher figure of merit. The naive ranking is dominated by vectors containing a *random*-quantile leg, with apparent gains up to $55\times$ (all of the global top 20 carry a random leg). We trace this not to the randoms being noise — their environment-classified *monopoles* carry real cosmic-web signal — but to their *quadrupoles*, which have no peculiar-velocity (RSD) signal and whose noisy derivatives inflate the Fisher; flagging the random legs is thus a conservative choice pending derivative-noise propagation. Among the clean *data*-leg vectors the gains are real: adding a single full $\times$ data-quantile cross buys $\sim 5\times$ in the three-parameter FoM, and a greedy forward selection that spans the four density quantiles reaches a five-stem vector (`full+xdQ1+dQ1+dQ2+xdQ3+dQ4`) with $\text{FoM}_3 = 43\times$ — marginalised errors tighter than the standard 2PCF by $3.4\times$ (ω_b), $4.1\times$ (ω_c), $3.6\times$ (n_s) and $2.8\times$ (σ_8), with clear diminishing returns past four stems. We document the method, the random-quadrupole diagnosis, the honest ranking, and the greedy chain.

1 Question and data

ASTRA splits a galaxy sample into four environment quantiles (Q1 underdense $\rightarrow$ Q4 overdense) and measures, per box, the multipoles $\xi_{0,2}(s)$ of: the full-sample auto-correlation (`tpcf_full_data`, our baseline), the four data-quantile autos (`dQ1..4`), the four random-quantile autos (`rQ1..4`), the four full $\times$ data-quantile crosses (`xdQ1..4`), and the four full $\times$ random-quantile crosses (`xrQ1..4`). That is $17 \text{ stems} \times \{\ell = 0, 2\} \times 15 \text{ bins per run}$.

The signal data are the $9 \times 50 = 450$ phase-matched full-box runs (`data/fullbox/`); cosmology derivatives come from the global linear response fit (`derivative_global_*`, condition number 1.34 on the full grid). The covariance of one $500 h^{-1}$ Mpc subbox is estimated from the mean-subtracted subboxes pooled across all nine cosmologies, $9 \times 64 = 576$ fluctuation samples, and scaled to the full $2000 h^{-1}$ Mpc box volume ($C/64$). The HOD nuisance covariance C_{HOD} is the empirical scatter of the 50 same-phase c000 draws.

2 Method

The analysis is implemented in `scripts/fisher_vector_search.py`, which reuses the assembly and Fisher machinery of `scripts/fisher_joint.py`. The steps are:

1. **Candidate vectors.** Each candidate is one or two ASTRA stems, taken at *native* 15-bin resolution in mono+quad ($\ell = 0, 2$). The 576-sample pooled covariance keeps the Hartlap factor benign even at $n_b = 60$ (pair) or 90 (pair added to the full auto), so no rebinning is needed. We evaluate the 16 singles and all $\binom{16}{2} = 120$ pairs.
2. **Two framings.** *Standalone* scores the candidate by itself against the full auto (“is it better than the 2PCF?”). *Additive* scores the full auto with the candidate appended (“what does ASTRA add to the 2PCF?”). The additive framing is the physical measure of marginal information and is our headline.
3. **Fisher and marginalisation.** For each candidate we build the 16×16 joint Fisher over the four cosmological parameters $\{\omega_b, \omega_c, n_s, \sigma_8\}$ and the twelve yuan23 HOD parameters, $F = D C^{-1} D^T$ with Hartlap-corrected precision, add a Gaussian yuan23 prior on the HOD block, and read off the HOD-marginalised 4×4 cosmology covariance $\text{Cov} = (F + F_{\text{prior}})_{[\text{cosmo}]}^{-1}$ (route a).
4. **Figure of merit.** We reduce Cov to

$$\text{FoM}_4 = [\det \text{Cov}_4]^{-1/2}, \quad \text{FoM}_3 = [\det \text{Cov}_3]^{-1/2}, \quad (1)$$

where Cov_3 drops σ_8 (the upper-left 3×3 block, already marginalised over σ_8). The *gain* of a candidate is $\text{FoM}/\text{FoM}_{\text{full auto}}$. We headline FoM_3 because the σ_8 derivative is unreliable (Sec. 4), so a four-parameter FoM inherits that unreliability.

3 Background: how the covariance and the derivatives are estimated

This section explains, in simple terms, the two main ingredients of the Fisher matrix in Eq. (4): the covariance C (how noisy the measurement is) and the derivatives D (how the measurement changes when a parameter changes). It is written for a reader who is new to the method.

The simulation boxes

All our data come from the AbacusSummit simulations. Each simulation is a cubic box with a side of $2000 h^{-1} \text{ Mpc}$ (about 3 billion light-years). Galaxies are placed inside the box with an HOD (halo occupation distribution) model, at redshift $z = 0.5$. We use nine different cosmologies. For each cosmology we have 50 galaxy catalogues, each one with different HOD parameters. So we have $9 \times 50 = 450$ boxes in total. All boxes use the same random “phase” (called **ph000**): they start from the same initial pattern of matter. This matters because, when we compare two boxes, the random differences between them (“cosmic variance”) then mostly cancel.

The covariance, from subboxes

The covariance tells us how much our measurement would change if we measured it again in a different patch of the Universe. Ideally we would use many separate big boxes, but big boxes are expensive to simulate. Instead we cut one full box into smaller cubes. We divide the $2000 h^{-1} \text{ Mpc}$ box into $4 \times 4 \times 4 = 64$ subboxes, each with a side of $500 h^{-1} \text{ Mpc}$. We measure the data vector in every subbox. The scatter among the 64 subboxes gives an estimate of the covariance of one $500 h^{-1} \text{ Mpc}$ subbox.

One box gives only 64 subboxes. That is not many compared with the length of our data vector (up to 180 numbers). To do better we combine the subboxes from all nine cosmologies. For each cosmology we first subtract its own mean, so that only the random fluctuations remain, and then we put all the subboxes together. This gives $9 \times 64 = 576$ samples. Here we assume the level of noise is about the same in every cosmology, which is a good approximation over our small range of cosmologies.

Our real measurement uses the full $2000 h^{-1}$ Mpc box, not a small subbox. A larger volume gives a smaller error. The variance goes down in proportion to the volume, so we divide the subbox covariance by 64 (the number of subboxes in a full box) to get the covariance of the full box.

Finally, a covariance estimated from a limited number of samples gives a slightly biased result when we invert it. We correct this with the standard ‘‘Hartlap factor’’, $(N - n_b - 2)/(N - 1)$, where $N = 576$ is the number of samples and n_b is the number of data points. For the five-stem vector ($n_b = 180$) this factor is 0.685. Using 576 samples instead of 64 is exactly what lets us keep the quadrupole at full resolution: with only 64 samples a 180-point vector could not be inverted at all (we would need $N > n_b$).

How robust is this covariance?

It is fair to ask how much we can trust a covariance built this way. We checked it in three ways, and the short answer is: it is robust enough for *comparing* data vectors (the main goal of this note), but the *absolute* error bars should be read with some caution.

The reassuring test is a ‘‘jackknife’’: we rebuild the covariance nine times, each time leaving one cosmology out of the pool ($8 \times 64 = 512$ samples), and recompute the five-stem FoM₃. The result changes by only about 6% (about 2% per parameter). So the answer does not depend on which cosmologies we pool, and because every data vector uses the *same* covariance, their ranking is trustworthy.

There are, however, three weaknesses that the jackknife does not see:

- **The subboxes are not fully independent.** The 64 subboxes come from a single box, so they share the largest waves and neighbouring subboxes are correlated. The true number of independent samples is therefore below 576. This means the real Hartlap correction should be a little stronger than the 0.685 we use, and the ratio of samples to data points (here $576/180 \approx 3.2$) is in practice even smaller. A ratio of only a few is modest: the inverse covariance is somewhat noisy, and indeed the correlation matrix has a large condition number ($\sim 10^4$), meaning some directions are close to unconstrained.
- **Open boundaries.** A $500 h^{-1}$ Mpc subbox has open edges and feels large-scale ‘‘super-sample’’ modes; the full $2000 h^{-1}$ Mpc box is periodic and does not. Dividing the subbox covariance by 64 assumes a simple volume scaling and ignores this difference, so the absolute covariance, especially on large scales, is only approximate.
- **Pooling assumption.** Pooling across cosmologies assumes the noise is the same for every cosmology. This is mild over our small parameter range, but it is an assumption.

In short, the *relative* results (which vector wins, by how much) are robust to roughly the 6% level; the *absolute* σ values should be treated as estimates good to perhaps 10–20%. The clean way to remove these approximations is to estimate the covariance from many *independent* full boxes — for example the 25 AbacusSummit phases (ph000–ph024) measured as periodic boxes, or a large suite of approximate mock catalogues — which would give many more than 576 truly independent

samples and drop the open-boundary problem entirely. That is the recommended next step before quoting absolute error bars.

The derivatives, from a response model

The Fisher matrix also needs the derivatives: how the data vector changes when we change each parameter by a small amount. We get these from a “global response model”. We fit one simple straight-line (linear) relation to all 450 boxes at the same time, separately for each bin of the data vector:

$$\xi = \xi_0 + \sum_p a_p \theta_{\text{cosmo},p} + \sum_q b_q \theta_{\text{HOD},q}, \quad (2)$$

where the θ_{cosmo} are the four cosmological parameters and the θ_{HOD} are the twelve HOD parameters. The slopes a_p are the cosmology derivatives; the slopes b_q are the HOD derivatives. Because the HOD terms are fitted at the same time, the cosmology slopes come out “clean”: they are not spoiled by the fact that different cosmologies happen to have different HOD values. And because all boxes share the same phase, cosmic variance cancels in the slopes. This fit is very well behaved on the full grid (its condition number is 1.34).

A more flexible fit: the Gaussian Process

The straight-line fit assumes the data vector changes in exact proportion to each parameter. The true response is only approximately linear. To test this we also used a Gaussian Process (GP). A GP is a flexible method that learns a smooth curve from data without assuming a straight line; as a bonus, it also gives an error bar on every prediction.

We trained one GP for each cosmology. The input was the twelve HOD parameters and the output was the data vector, using that cosmology’s 50 catalogues as training points. The HOD derivative is then the slope of the GP curve at the central HOD point. Two things came out of this test. First, the HOD response is mildly nonlinear: the GP fits the data clearly better than the straight line (it captures about 75% of the variation, against about 18% for the line). Second, only a few HOD parameters — mainly the typical mass of the host halo and its scatter — drive most of the change; the other parameters matter little.

The headline numbers in this note use the simpler linear response model, because it is stable and well understood on the full grid. The GP is a refinement and a cross-check; turning it into final derivatives that also carry their own error bars is left for future work. (The same GP idea, applied separately in each cosmology and evaluated at one common HOD point, is also how we can build “HOD-matched” derivatives, but that is beyond the scope of this note.)

4 The random-quadrupole trap

Table 1 and the bright **rQ4/xrQ4/xrQ1** band in Fig. 2 show that the naive sweep is dominated by vectors containing a random leg, with implausibly large gains (FoM₃ up to 55×). Taken at face value this would say the random-quantile statistics are the most powerful of all, which deserves scrutiny. The key is to separate the monopole from the quadrupole; the trap is the *quadrupole*, not the random statistic as a whole.

- **The random-quantile monopoles carry real signal.** Once the uniform random points are classified by their Delaunay environment, the overdense quantile genuinely sits in overdense regions and the underdense one in voids. Their auto-correlation therefore measures the

Table 1: Raw top of the additive ranking (full auto + stem(s)). A star marks a candidate that contains a random leg (rQ or xrQ). *Every* entry in the global top eight, and all 20 of the global top 20, carry a random leg, despite random legs underperforming in every earlier single-parameter test — a warning sign examined in the text.

	vector	FoM ₃ gain	FoM ₄ gain
singles	xrQ4*	12.2	13.5
	rQ4*	8.6	9.9
	xrQ1*	7.8	8.6
pairs	xrQ1+xrQ4*	55.8	64.6
	xdQ1+xrQ4*	44.6	59.0
	xdQ4+xrQ4*	43.4	60.3

clustering of the cosmic-web *environments* themselves — a real, cosmology-dependent quantity, not Poisson noise. Indeed the extreme random monopoles are as signal-rich as the data quantiles (rQ1, rQ4 $\ell 0$ noise/signal ≈ 0.05 – 0.09).

- **The random-quantile quadrupoles are physically weak.** The quadrupole of the *data* comes from peculiar velocities (redshift-space distortions). The random points have *no* peculiar velocities, so their quadrupole carries almost no real signal: the little anisotropy present is a geometric byproduct of the classification and is dominated by realisation noise (rQ2/3 $\ell 2$ noise/signal ≈ 0.7 – 0.8).
- **The inflation is derivative noise in the quadrupole.** A Fisher matrix treats the derivative D as exact; noise in D that is not represented in the covariance C inflates $D C^{-1} D^T$, so the Fisher reads derivative noise as constraining power. A mono+quad random vector carries a signal-rich monopole *and* a noise-limited quadrupole, and it is the noisy quadrupole derivative that drives the spurious gains. The signature fits: a random vector is barely better than the full auto on its own (xrQ4 standalone FoM₃ = 1.6) yet explodes when appended.

So the caution is narrower than “the randoms are noise”. Their monopoles are genuine cosmic-web environment signal; it is the velocity-free quadrupoles, with their noisy derivatives, that inflate the Fisher — and, because the randoms trace the *same* density field as the data, large *independent* gains are in any case implausible on redundancy grounds. Flagging every random-leg candidate is therefore a *conservative* choice, not a proof that they are worthless: it removes the quadrupole-noise inflation at the price of also setting aside the real random-monopole information. The clean resolution is to propagate the (emulator) derivative variance, which would deflate the noisy quadrupole derivatives while keeping the genuine monopole signal (future work). Until then we base the physical conclusions on the data legs only.

5 The honest, data-leg ranking

Restricting to data-leg candidates (Table 2, Fig. 1) gives a clean and physically sensible picture:

- **Singles.** The full×data-quantile crosses at the extremes lead — xdQ1 (void) and xdQ4 (knot), FoM₃ gain ≈ 5 — followed by the extreme data autos dQ4, dQ1 (≈ 4). The middle quantiles (Q2, Q3) are weaker. Each of these improves ω_b and n_s by ~ 1.6 – $1.7\times$ over the full auto. Note

Table 2: Data-leg candidates only (no random leg). Additive framing, HOD-marginalised. Baseline full-auto errors: $\sigma(\omega_b) = 5.7 \times 10^{-4}$, $\sigma(\omega_c) = 1.7 \times 10^{-3}$, $\sigma(n_s) = 1.3 \times 10^{-2}$, $\sigma(\sigma_8) = 1.4 \times 10^{-2}$.

	vector	FoM ₃ gain	$\sigma(\omega_b)$	$\sigma(\omega_c)$	$\sigma(n_s)$
singles	xdQ1	5.1	3.3×10^{-4}	8.4×10^{-4}	8.3×10^{-3}
	xdQ4	5.0	3.2×10^{-4}	8.4×10^{-4}	8.5×10^{-3}
	dQ4	4.1	3.3×10^{-4}	9.8×10^{-4}	8.4×10^{-3}
	dQ1	3.9	4.2×10^{-4}	1.1×10^{-3}	7.7×10^{-3}
pairs	dQ1+xdQ1	14.0	2.8×10^{-4}	—	6.0×10^{-3}
	dQ2+xdQ1	12.8	2.5×10^{-4}	—	4.9×10^{-3}
	xdQ3+xdQ4	12.2	2.6×10^{-4}	—	5.0×10^{-3}

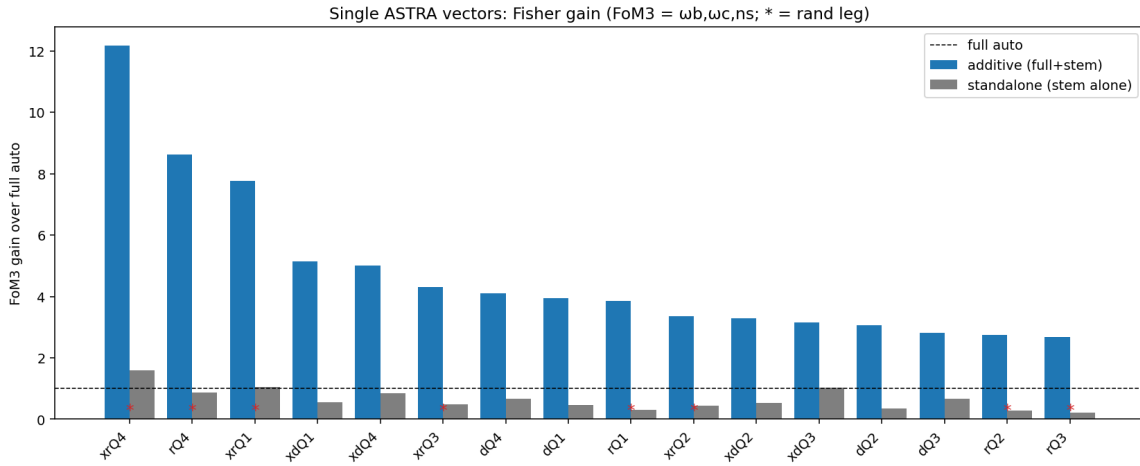


Figure 1: Single-vector FoM₃ gain over the full auto, additive (blue) and standalone (grey), ranked. Red stars mark random-leg vectors, whose additive gains are inflated by noise in their velocity-free quadrupole derivatives (Sec. 4). Every vector is at or below the full-auto line on its own; the gain is degeneracy-breaking, realised only when appended.

that on its own each vector is *worse* than the full auto (standalone FoM₃ < 1.1); the gain is entirely degeneracy-breaking when added — the ASTRA thesis.

- **Pairs.** The best clean pair is **dQ1+xdQ1** (void data auto + void data cross), FoM₃ gain 14, i.e. a factor ~ 2.0 on ω_b and ~ 2.2 on n_s over the full auto. Combinations of an extreme data auto with an extreme data cross dominate; pairs of two middle quantiles, or two stems from the same family, add little (redundant).
- **σ_8 is untouched.** No vector improves σ_8 appreciably ($1.4 \times 10^{-2} \rightarrow \sim 1.0 \times 10^{-2}$), consistent with σ_8 being amplitude-degenerate with the HOD and with its derivative being unreliable. This is why FoM₃ (which drops σ_8) is the trustworthy score; Fig. 3 confirms the per-parameter picture.

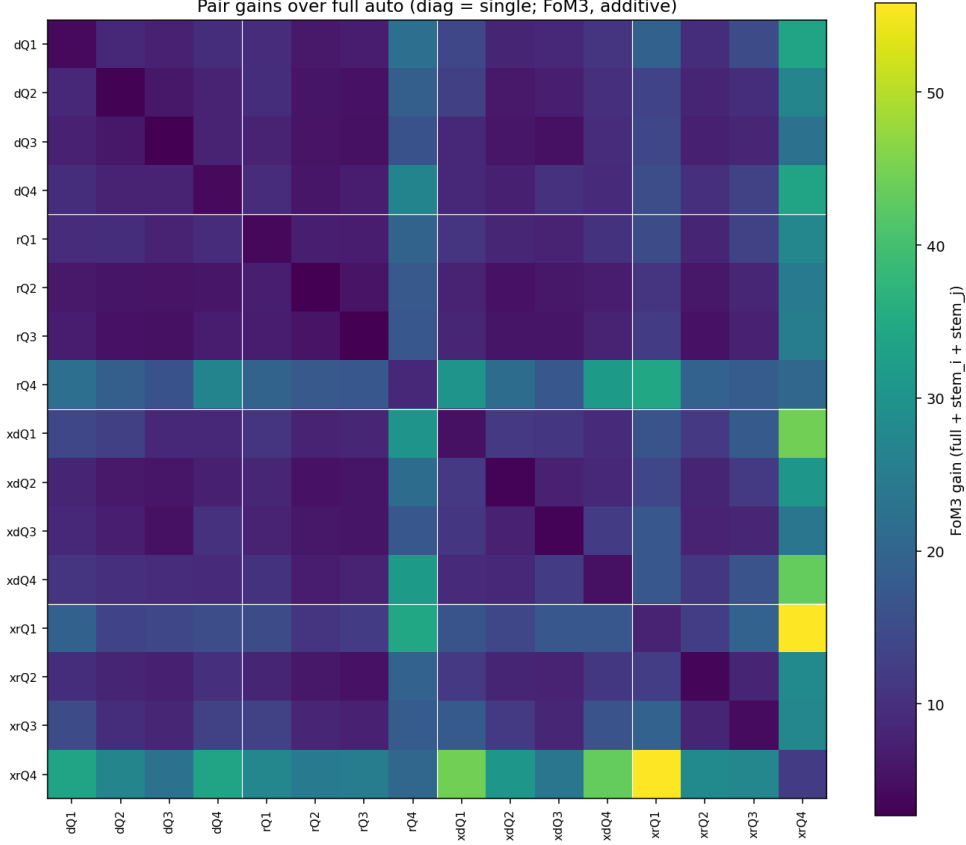


Figure 2: Additive FoM₃ gain for every pair, full auto + stem_i + stem_j (diagonal = single). White lines separate the four families (dQ, rQ, xdQ, xrQ). The bright band on the rQ4/xrQ4/xrQ1 rows and columns is driven by the noisy random-quadrupole derivatives (Sec. 4); the data-leg quadrants (dQ, xdQ) show the modest, trustworthy gains.

6 Greedy forward selection: how many stems, and which?

The single- and pair-level results raise the obvious question: if one extreme data cross already buys $\sim 5\times$ and the void auto+cross pair buys $14\times$, how far does it go? We answer it with a *greedy forward selection* restricted to the eight data-leg stems (`scripts/fisher_greedy_chain.py`). Starting from the full auto, at each step we append the single data-leg stem that most increases FoM₃, keep it, and repeat, up to five added stems. Greedy selection is not guaranteed to find the globally optimal subset, but it is the natural way to read off the *marginal* value of each successive statistic and to expose redundancy.

The chain is full+xdQ1+dQ1+dQ2+xdQ3+dQ4 (Table 3, Fig. 4). Three things stand out.

- **Strong diminishing returns.** The per-step gain falls $5.1 \rightarrow 2.7 \rightarrow 1.8 \rightarrow 1.3 \rightarrow 1.3$. The first two stems carry most of the information (FoM₃ reaches 14 of the eventual 43 by step two); beyond step four each new statistic adds only $\sim 30\%$, while the bin count and Hartlap/redundancy cost keep rising. A four-stem vector is the practical sweet spot.
- **Gains come from spanning environments, not repetition.** Greedy does not pile up void statistics; after the void auto+cross it reaches for a *different* density bin (dQ2), then mid-high (xdQ3), then the knot (dQ4), touching all four quantiles by step five. Different environments

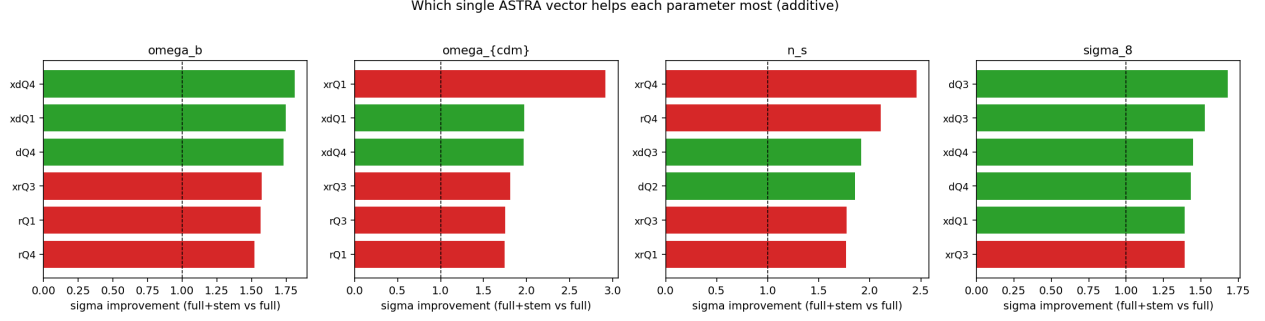


Figure 3: For each cosmological parameter, the single ASTRA vectors that most improve its marginalised error when added to the full auto. Green bars are data-leg vectors (trustworthy); red bars are random-leg vectors, whose gains lean on the noisy velocity-free quadrupole and are not yet trustworthy.

Table 3: Greedy chain over the data-leg stems. “marg.” is the multiplicative gain in FoM_3 from that one step. σ values are the HOD-marginalised errors of the cumulative vector. The chain visits all four density quantiles by step five.

step	add	meaning	FoM_3	marg.	$\sigma(\omega_b)$	$\sigma(n_s)$	$\sigma(\sigma_8)$
0	—	full auto	1.0	—	5.7×10^{-4}	1.3×10^{-2}	1.4×10^{-2}
1	xdQ1	void cross	5.1	5.1	3.3×10^{-4}	8.3×10^{-3}	1.0×10^{-2}
2	dQ1	void auto	14.0	2.7	2.8×10^{-4}	6.0×10^{-3}	7.5×10^{-3}
3	dQ2	low-mid auto	24.7	1.8	2.3×10^{-4}	4.3×10^{-3}	6.8×10^{-3}
4	xdQ3	mid-high cross	32.5	1.3	2.1×10^{-4}	3.8×10^{-3}	5.4×10^{-3}
5	dQ4	knot auto	43.3	1.3	1.7×10^{-4}	3.6×10^{-3}	5.0×10^{-3}

respond to cosmology and to the HOD in different ways, so each new environment breaks a *complementary* degeneracy; a second statistic from the same environment would be largely redundant.

- **σ_8 moves only in combination.** No single vector improved σ_8 (Sec. 5), but the cumulative five-stem vector tightens it $2.8 \times (1.4 \times 10^{-2} \rightarrow 5.0 \times 10^{-3})$. The multi-environment combination partially lifts the amplitude degeneracy that defeats any one statistic — though σ_8 remains the least trustworthy entry because its derivative is unreliable, and FoM_3 (which excludes it) stays our headline score.

The five-stem vector versus the 2PCF

Figure 5 is the head-to-head Fisher comparison of the standard full-sample 2PCF (`tpcf_full_data`, mono+quad) and the five-stem ASTRA vector, both HOD-marginalised, generated by `scripts/fisher_compare.f`. The five-stem 68/95% contours sit nested well inside the full-auto contours on every panel, with the marginalised errors tighter by

$$\sigma(\omega_b) \ 3.4\times, \quad \sigma(\omega_c) \ 4.1\times, \quad \sigma(n_s) \ 3.6\times, \quad \sigma(\sigma_8) \ 2.8\times, \quad (3)$$

i.e. $\text{FoM}_3 = 43\times$. Crucially the blue ellipses are not merely smaller but *rotated* relative to grey: the environment splits change the orientation of the parameter degeneracies, which is the geometric

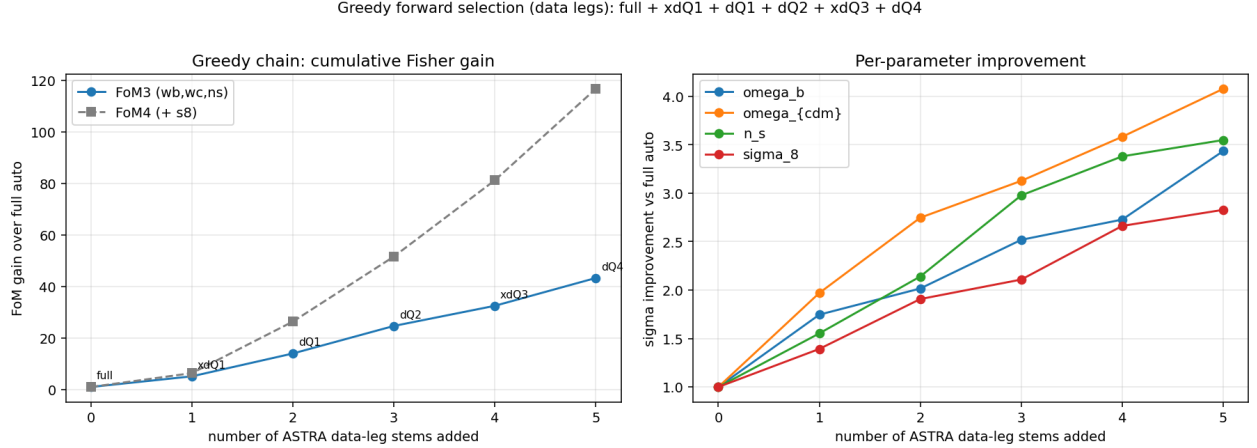


Figure 4: Greedy forward selection over the data legs. Left: cumulative FoM₃ (and FoM₄) gain over the full auto versus the number of ASTRA stems added — the curve flattens after ~ 4 stems. Right: per-parameter σ improvement over the full auto. Labels mark the stem chosen at each step.

signature of genuinely new information rather than a uniform rescaling. The five-stem 95% region is inside the full-auto 68% region for most parameter pairs — a qualitatively different constraint from a single $2 h^{-1}$ Mpc box, by reanalysis alone.

7 Anatomy of the 5-stem Fisher computation

This section opens up the Fisher calculation behind Fig. 5 step by step, so every ingredient is visible. The five-stem vector concatenates six pieces (the full auto plus the five selected stems), each in mono+quad at the native 15-bin resolution, for a data vector of $n_b = 6 \times 2 \times 15 = 180$ bins. All figures are produced by `scripts/fisher_5stem_details.py`.

The Fisher matrix is

$$F = D C^{-1} D^T, \quad (4)$$

where D is the $(4 + 12) \times n_b$ matrix of derivatives of the data vector with respect to the four cosmological and twelve HOD parameters, and C is the data covariance. The cosmology constraints follow from inverting either the cosmology block alone (conditional, HOD fixed) or the full matrix plus the HOD prior and taking the cosmology sub-block (marginalised). The four steps below are the four factors in Eq. (4).

Step 1 — the data vector

Figure 6 shows the fiducial (c000 ensemble-mean) data vector, one panel per piece. The monopole amplitude grows monotonically from the void (dQ1) to the knot (dQ4, an order of magnitude larger): the overdense quantile is the most strongly clustered and highest-bias tracer, the underdense one the least. The quadrupoles ($\ell = 2$) carry the redshift-space anisotropy. These six curves, stacked $[\ell 0, \ell 2]$ per piece, are the 180-component vector whose information content we are dissecting.

Step 2 — the derivatives D

Figure 7 shows the four cosmology rows of D (the global-response derivatives $\partial \xi / \partial \theta$) across all 180 bins, with the six piece blocks marked. Two features matter for the Fisher information:

- The derivative amplitude is largest on the high-bias **dQ4** piece, especially at small s — the knot quantile is the most cosmology-responsive.
- The *shape* of the response differs from piece to piece (e.g. the ω_b acoustic feature versus the broadband n_s tilt appear with different relative weights in each environment). It is precisely this piece-to-piece difference in $\partial\xi/\partial\theta$ that lets the environment splits break parameter degeneracies the full auto cannot.

(The twelve HOD rows of D , not shown, enter the marginalisation in Step 4.)

Step 3 — the covariance C

Figure 8 is the 180×180 correlation matrix of the data vector, estimated from the mean-subtracted subboxes pooled across all nine cosmologies (576 samples) and scaled to the full-box volume. The block structure is the crux of how much *independent* information the vector holds:

- Within each piece, neighbouring s -bins are positively correlated, and the monopole and quadrupole blocks are cross-correlated.
- Across pieces, the full auto is strongly correlated with the full $\times$ data crosses (**xdQ1,xdQ3**) because they literally share the full-sample leg, and the quantile autos are mutually correlated through the shared density field. These off-diagonal correlations are why the gains saturate: the pieces are not independent, and C^{-1} correctly down-weights the redundancy.

The 576 pooled samples are what make $n_b = 180$ affordable: the Hartlap factor is $(576 - 180 - 2)/(576 - 1) = 0.685$, comfortably positive, where the 64-subbox estimate would have failed ($n_b > 64$).

Step 4 — the Fisher matrix and parameter covariance

Combining D and C via Eq. (4) with the Hartlap-corrected precision gives the parameter covariance. Figure 9 shows the 4×4 cosmology parameter correlation matrices, conditional (HOD fixed) and marginalised (HOD nuisances marginalised with the yuan23 prior), with the per-parameter σ in the titles:

	$\sigma(\omega_b)$	$\sigma(\omega_c)$	$\sigma(n_s)$	$\sigma(\sigma_8)$
conditional (HOD fixed)	2.9×10^{-5}	2.0×10^{-4}	7.6×10^{-4}	2.9×10^{-3}
marginalised	1.7×10^{-4}	4.1×10^{-4}	3.6×10^{-3}	5.0×10^{-3}

Two lessons:

- **Marginalising over the HOD costs a factor few** per parameter ($6\times$ on ω_b , $5\times$ on n_s): the cosmology directions that overlap the HOD response are degraded most. This is the honest price of not knowing the galaxy–halo connection, and it is the regime all our headline numbers are quoted in.
- **Marginalisation re-shapes the degeneracies.** Conditionally, ω_b is strongly anti-correlated with σ_8 (-0.84) and ω_c with n_s (-0.79); after HOD marginalisation these soften and new correlations appear ($\omega_b\text{--}\omega_c = +0.55$, $n_s\text{--}\sigma_8 = -0.42$). The rotation of the ellipses in Fig. 5 is exactly this change of correlation structure.

Table 4: Multipole configuration comparison, HOD-marginalised. FoM₃ gain is relative to the standard 2PCF (full auto, mono+quad). The quadrupole is weak alone but adds substantially when combined, and is the main carrier of σ_8 information.

vector	config	n_b	FoM ₃ gain	$\sigma(\sigma_8)$
full auto	mono ($\ell 0$)	15	0.41	3.6×10^{-2}
full auto	quad ($\ell 2$)	15	0.12	4.0×10^{-2}
full auto	mono+quad	30	1.00	1.4×10^{-2}
5-stem	mono ($\ell 0$)	90	12.1	1.2×10^{-2}
5-stem	quad ($\ell 2$)	90	2.6	1.0×10^{-2}
5-stem	mono+quad	180	43.3	5.0×10^{-3}

8 Monopole, quadrupole, or both?

The vectors above use both multipoles; does the quadrupole earn its place? We test this two ways with `scripts/fisher_multipole_compare.py`: a configuration comparison (mono-only, quad-only, mono+quad) for both the full auto and the five-stem vector, and a greedy “poll” that treats each (piece, multipole) as one of twelve selectable units and records whether the monopole or quadrupole is chosen at each step (Fig. 10).

- **The monopole is the workhorse.** On its own it carries most of the information — 0.41 of 1.0 for the full auto, 12 of 43 for the five-stem vector. The quadrupole alone is weak (0.12 and 2.6).
- **But the quadrupole is not redundant.** Adding it to the monopole multiplies FoM₃ by $2.4\times$ for the full auto ($0.41 \rightarrow 1.0$) and $3.6\times$ for the five-stem vector ($12 \rightarrow 43$). Crucially it is the main source of σ_8 information: mono-only $\sigma(\sigma_8)$ improves $2.4\times$ when the quadrupole is added ($1.2 \times 10^{-2} \rightarrow 5.0 \times 10^{-3}$ for the five-stem vector) — the redshift-space growth signal the isotropic monopole cannot see.
- **The quadrupole matters more for ASTRA than for the 2PCF** ($3.6\times$ vs $2.4\times$): environment-dependent velocity fields give the quantile quadrupoles extra independent RSD information.
- **The greedy poll confirms the hierarchy.** Of the first six selected (piece, multipole) units, four are monopoles and two are quadrupoles; greedy front-loads the information-dense monopoles but reaches a quantile quadrupole (**dQ2- $\ell 2$**) as early as step four, so certain quadrupoles carry strong *independent* information.

The verdict is to use mono+quad: the monopole is primary, the quadrupole is complementary (mostly σ_8 and n_s), and its value is larger in the environment splits than in the plain 2PCF. This refines the earlier impression that “the quadrupole adds almost nothing” — that held for ω_b specifically and under a Hartlap-limited bin budget; with the pooled covariance the quadrupole clearly pays off.

9 Conclusions

1. The systematic sweep of all 16 singles and 120 pairs is dominated, at face value, by vectors with a random-quantile leg (gains up to $55\times$). The cause is specifically the random **quadrupole**:

the environment-classified random *monopoles* carry real cosmic-web signal, but the random quadrupoles have no peculiar-velocity (RSD) signal and their noisy derivatives inflate the Fisher. Flagging the random legs is a conservative choice (it also sets aside the real random-monopole information); any ranking here must noise-correct the random quadrupole, e.g. by propagating the derivative variance, before trusting those gains.

2. Among the **trustworthy data legs**, the full×data-quantile crosses at the density extremes (**xdQ1**, **xdQ4**) are the best single additions (FoM₃ gain ≈ 5), and the best clean pair is the void data auto with the void data cross, **dQ1+xdQ1** (FoM₃ gain 14, $\sim 2.4\times$ per parameter on ω_b, ω_c, n_s). The gains are pure degeneracy breaking: each vector is worse than the full auto alone.
3. A **greedy chain** over the data legs (full+xdQ1+dQ1+dQ2+xdQ3+dQ4) reaches FoM₃ = $43\times$ at five stems, tightening the standard-2PCF errors by $3.4\text{--}4.1\times$ on ω_b, ω_c, n_s and $2.8\times$ on σ_8 (Fig. 5). Returns diminish sharply ($5.1 \rightarrow 2.7 \rightarrow 1.8 \rightarrow 1.3$ per step), so ~ 4 stems is the practical optimum, and the gains come from *spanning* the four environments rather than repeating one.
4. Single vectors do not improve σ_8 (HOD-amplitude degeneracy, unreliable derivative), but the multi-environment combination does so modestly; FoM₃ excluding σ_8 remains the trustworthy score.
5. **Use both multipoles.** The monopole is the workhorse and the quadrupole is weak alone, but adding the quadrupole multiplies FoM₃ by $2.4\text{--}3.6\times$ and is the dominant source of σ_8 information; its value is larger for the ASTRA splits than for the plain 2PCF (Sec. 8).
6. **Next steps.** (i) Propagate the GP emulator derivative variance (calibrated with the $\sim 1.25\times$ inflation measured separately) so the noisy random-quadrupole derivatives are deflated and the random legs can be ranked honestly — this should recover the genuine random-monopole information; (ii) use the higher-iteration runs to denoise the random-quadrupole derivatives directly; (iii) revisit the four-parameter FoM once a trustworthy σ_8 derivative is available; (iv) estimate the covariance from many independent full boxes (e.g. the 25 AbacusSummit phases) before quoting absolute error bars — the relative rankings here are robust to the $\sim 6\%$ level (Sec. 3), but the absolute σ rely on the subbox approximation.

All numbers are reproduced by `scripts/fisher_vector_search.py` (the full sweep; ranked table in `plots/vector_search/vector_search_results.csv`), `scripts/fisher_greedy_chain.py` (the greedy chain), and `scripts/fisher_compare_full_vs_5stem.py` (the corner comparison), and `scripts/fisher_5stem_details.py` (the step-by-step anatomy of Sec. 7), and `scripts/fisher_multipole_compare.py` (the monopole/quadrupole study of Sec. 8).

Fisher 68/95%: full-auto 2PCF vs 5-stem ASTRA (full + xdQ1 + dQ1 + dQ2 + xdQ3 + dQ4), HOD-marginalised

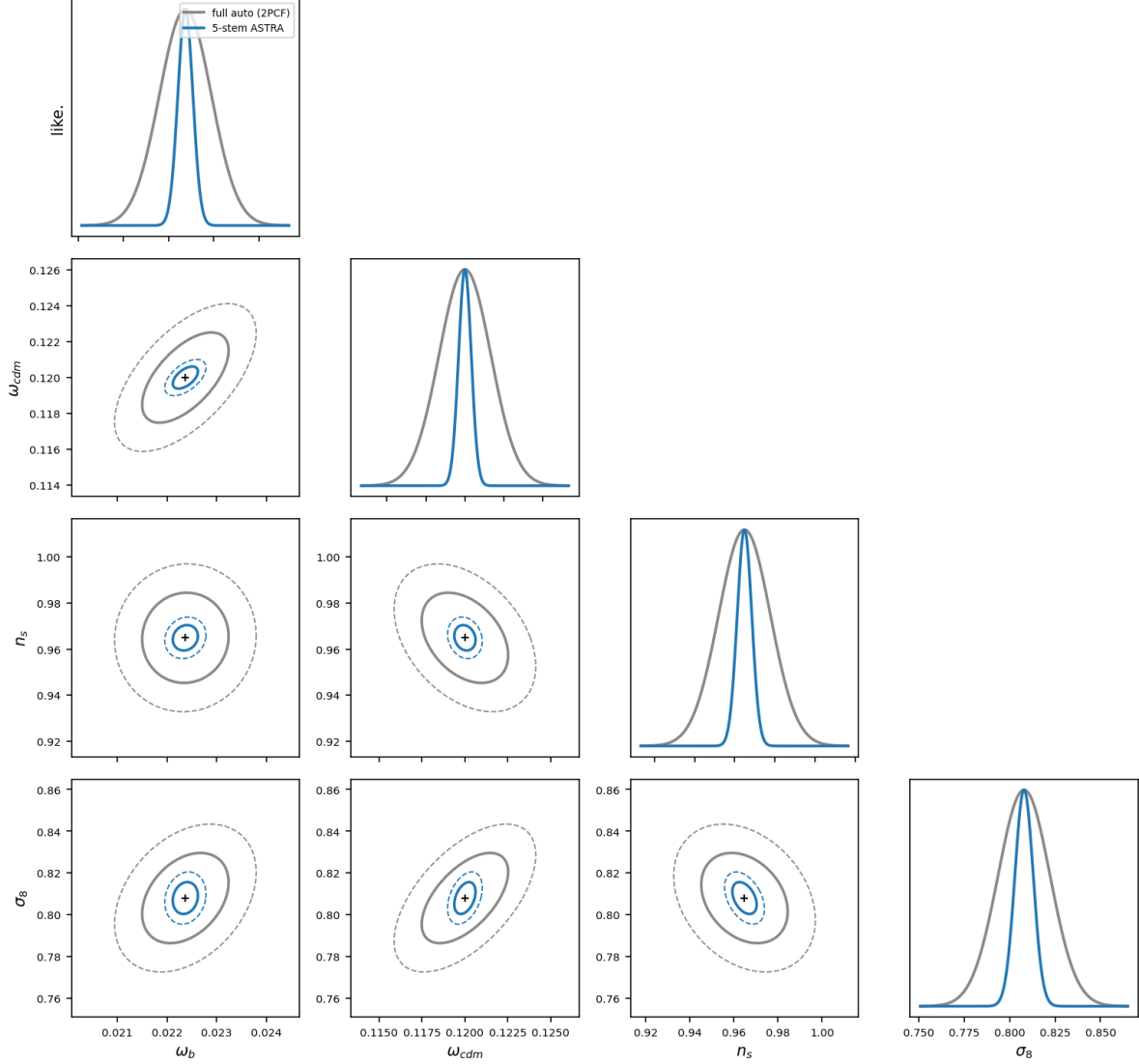


Figure 5: HOD-marginalised Fisher constraints: full-auto 2PCF (grey) versus the five-stem ASTRA vector **full+xdQ1+dQ1+dQ2+xdQ3+dQ4** (blue), 68% (solid) and 95% (dashed). The blue contours are both smaller and rotated — ASTRA shrinks and re-orientes the cosmology degeneracies.

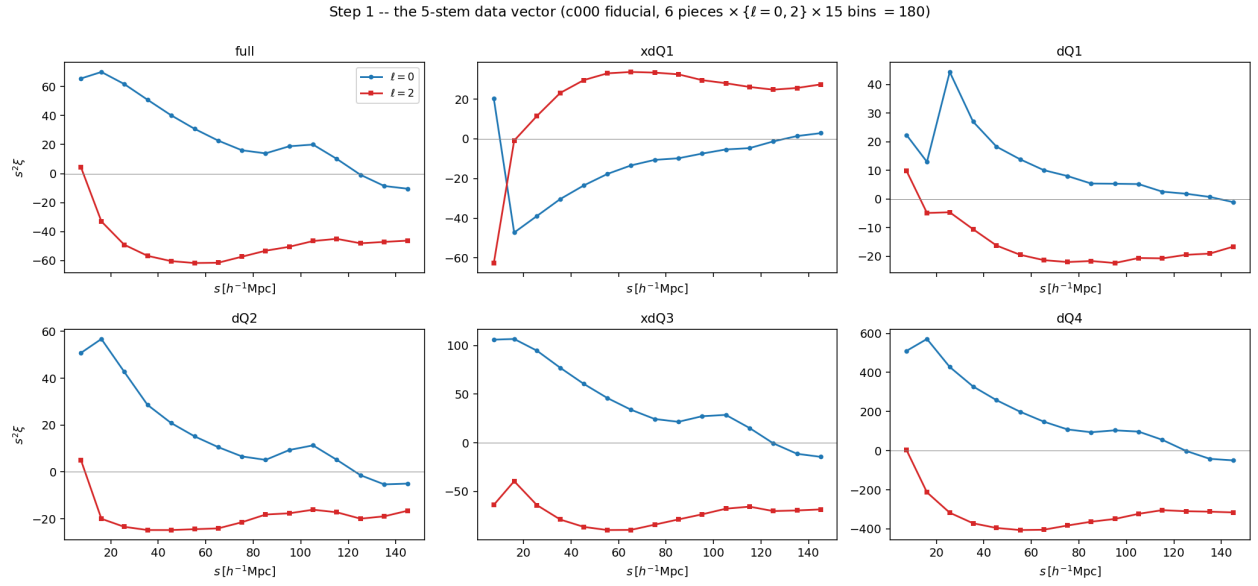


Figure 6: Step 1: the fiducial five-stem data vector, $s^2 \xi_{0,2}(s)$ per piece. Amplitude grows void (dQ1) $\rightarrow$ knot (dQ4).

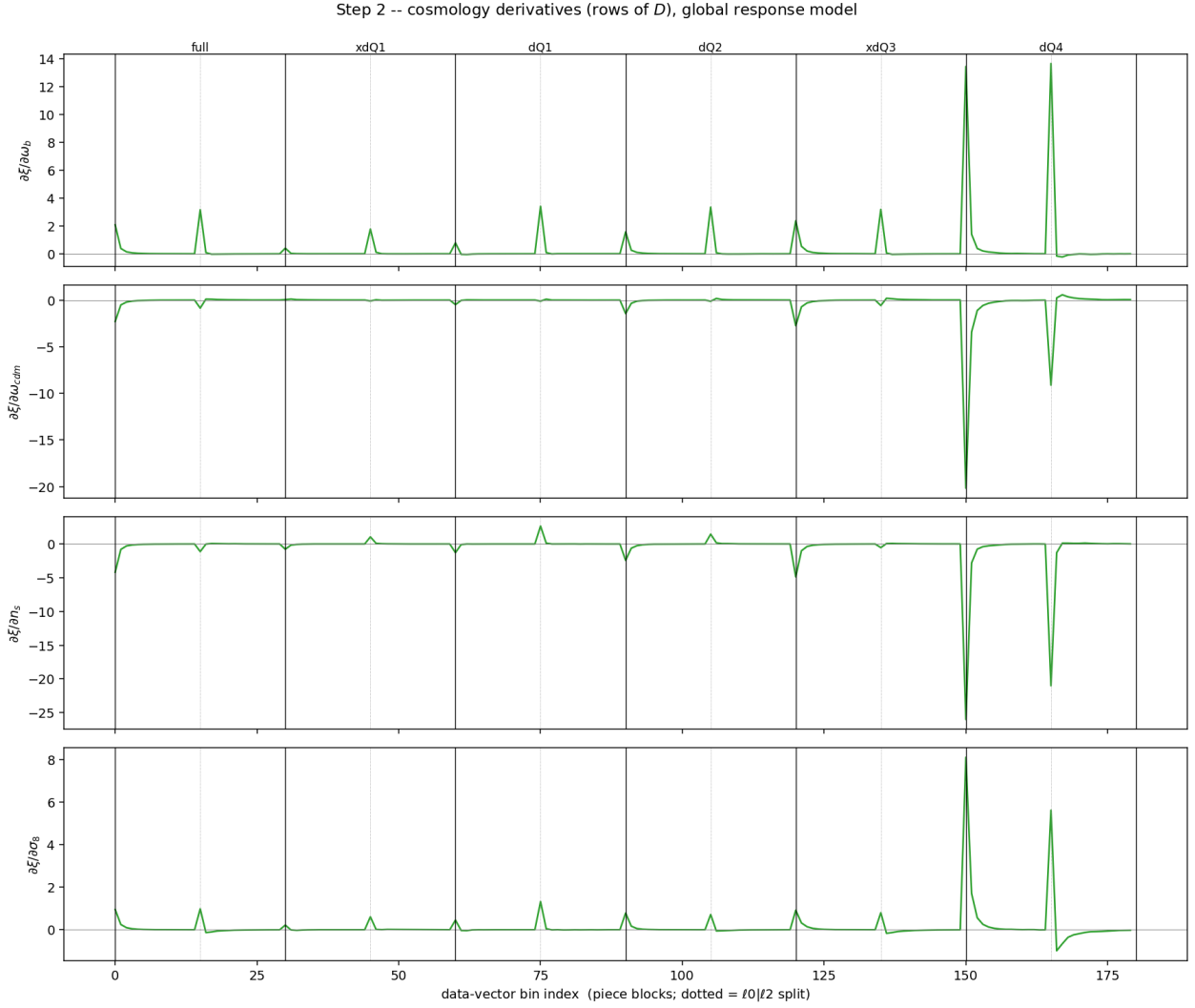


Figure 7: Step 2: the four cosmology rows of D , $\partial\xi/\partial\theta$, across the 180-bin vector. Solid lines separate the six pieces; dotted lines mark the ℓ_0/ℓ_2 split within each piece. The high-bias **dQ4** piece carries the largest response.

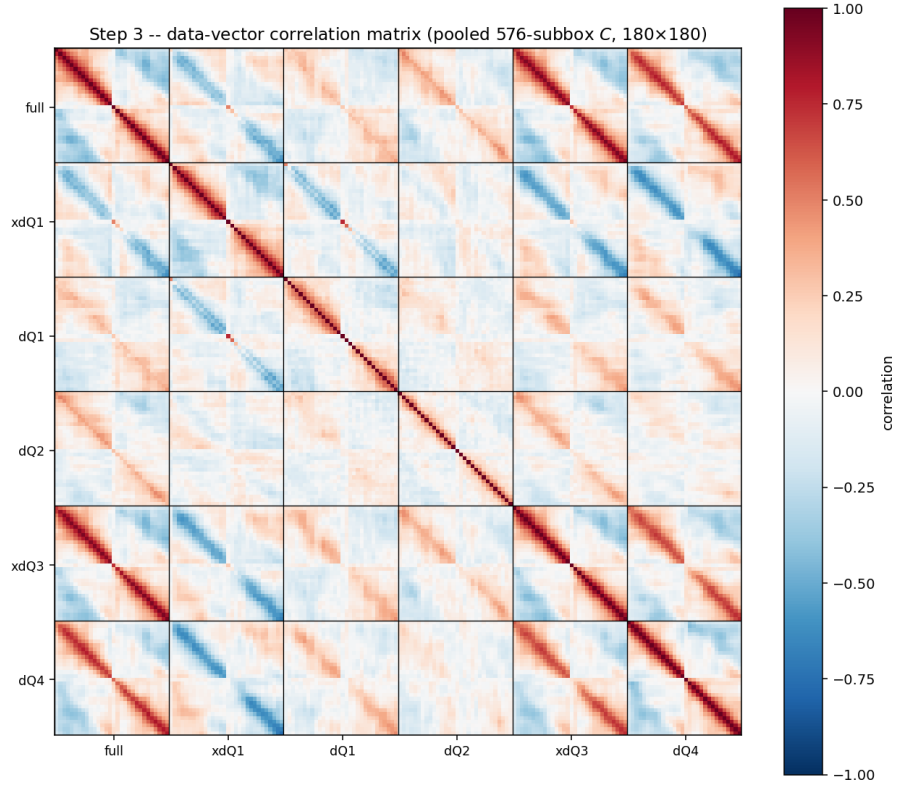


Figure 8: Step 3: the 180×180 data-vector correlation matrix (pooled 576-subbox C). Black lines separate the six pieces. Strong cross-piece correlations (e.g. full with the xd crosses) are the redundancy that limits the cumulative gain.

Step 4 -- cosmology parameter correlations from $F = DC^{-1}D^T$ (Hartlap 0.69)

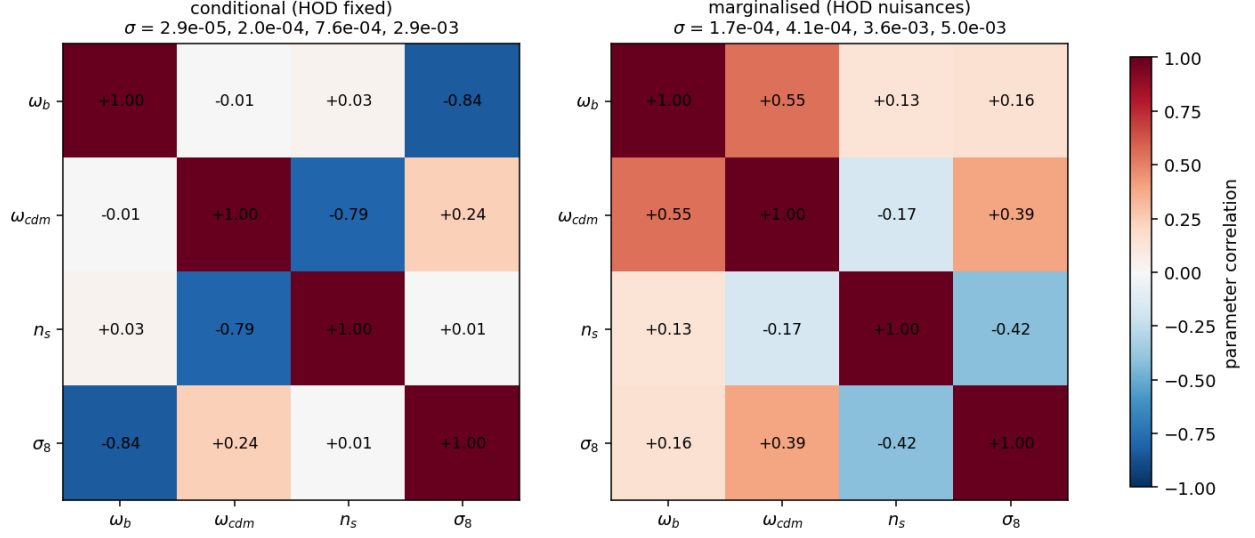


Figure 9: Step 4: cosmology parameter correlation matrices from $F = DC^{-1}D^T$, conditional (left) and HOD-marginalised (right), with the per-parameter σ in the titles. HOD marginalisation both inflates the errors and re-shapes the degeneracies.

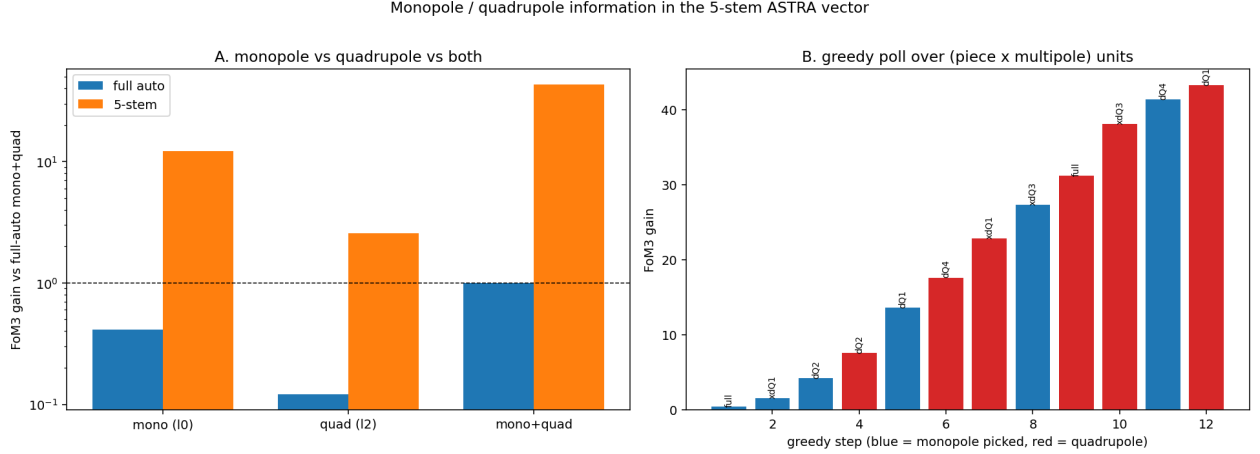


Figure 10: Left: FoM₃ gain (log scale) for mono-only, quad-only and mono+quad, for the full auto and the five-stem vector. Right: greedy poll over the twelve (piece, multipole) units; blue bars are monopole picks, red are quadrupole picks, with the cumulative FoM₃ at each step.